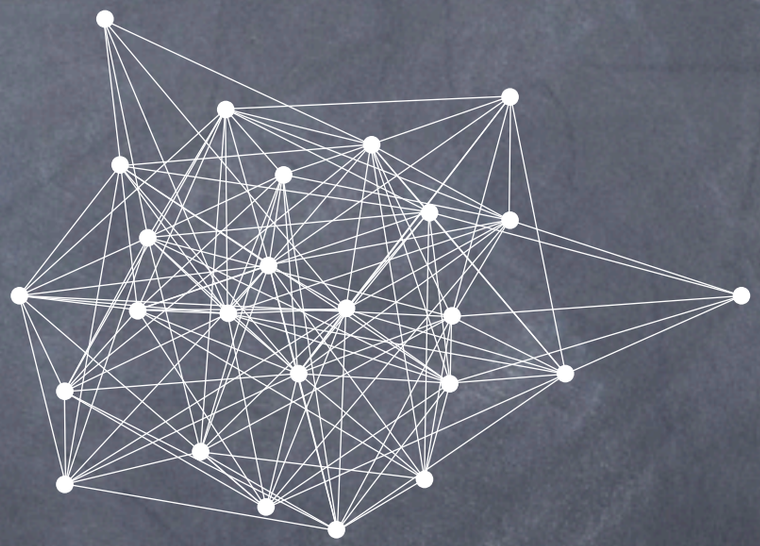
Graph Isomorphism Beyond the Hidden Subgroup Problem





$$\hat{R}^{(m)} = \{B_\Delta R_i B_\Delta^\dagger | R_i \in \mathcal{R}^{(m)}, B_\Delta R_i B_\Delta^\dagger \neq 0\}$$

Dave Bacon
University of Washinqton

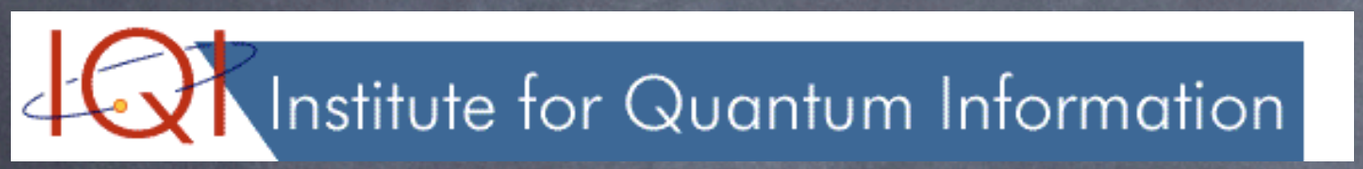
Happy Star Wars Day!



Namespace Overload





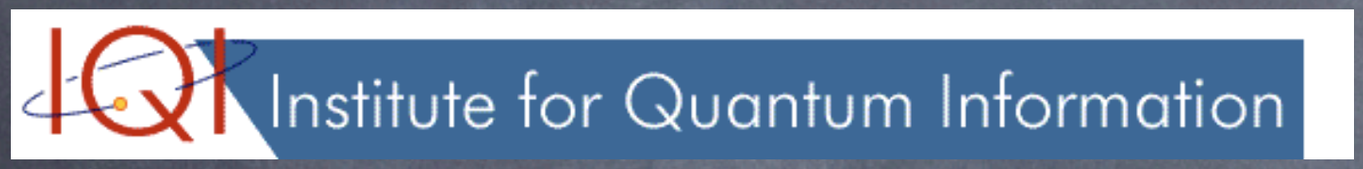




Namespace Overload











Institute for Extreme Quantum Computing
Institute for Extreme Quantum Information
Institute for Extreme Quantum Information Science

Namespace Overload

IQC Institute for Quantum Computing

IQSE Institute for Quantum Science and Engineering

IQI Institute for Quantum Information

Institute for Quantum Information Science at the University of Calgary

W. M. KECK FOUNDATION
CENTER FOR EXTREME QUANTUM INFORMATION THEORY
at the Massachusetts Institute of Technology

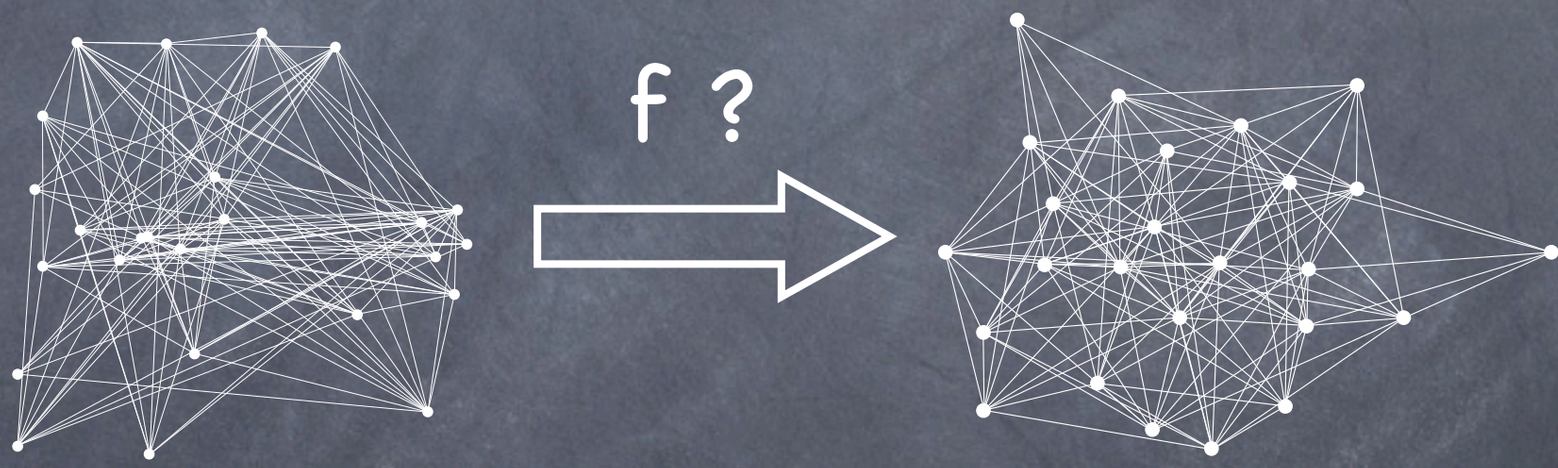
Institute for Extreme Quantum Computing
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Institute for Extreme Quantum Information Science

UNIVERSITY OF MARYLAND
jqi Joint Quantum Institute



Graph Isomorphism

GI: Given two graphs $G_1=(V_1,E_1)$ and $G_2=(V_2,E_2)$, does there exist a bijection, f, from vertex set V_1 to vertex set V_2, such that (u,v) is in E_1 iff (f(u),f(v)) is in E_2?



GA (automorphism): Given a graph G=(V,E), does there exists a non-trivial bijection from V to V such that (u,v) in E iff (f(u),f(v)) in E?

$$GA = GI_{rigid} \subseteq GI = GA_{search}$$



Goldilocks



GI is unlikely to be NP-complete as this would collapse the polynomial hierarchy to the second level.



Goldilocks



GI is unlikely to be NP-complete as this would collapse the polynomial hierarchy to the second level.



not Goldilocks



Goldilocks

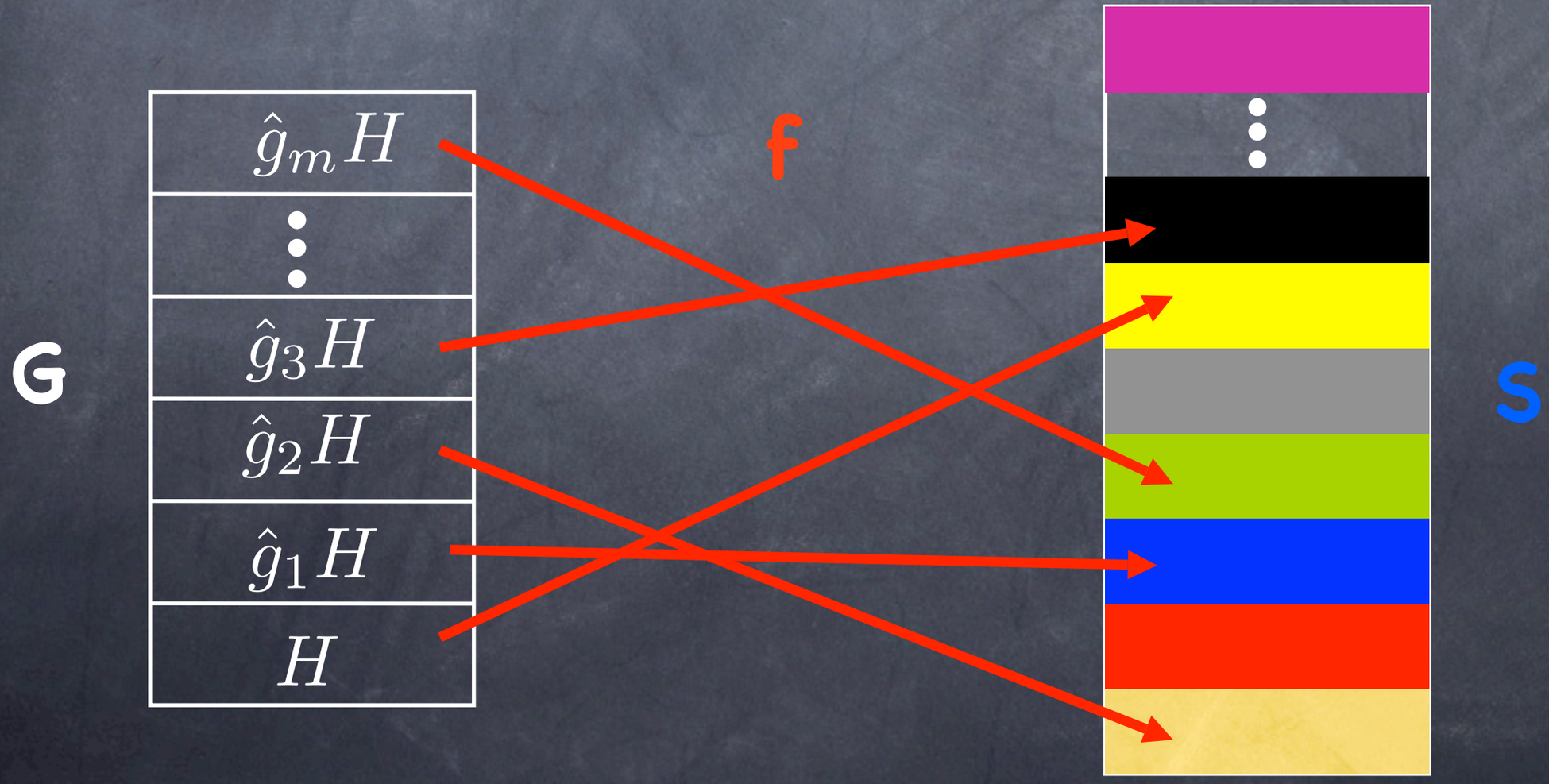


GI is unlikely to be NP-complete as this would collapse the polynomial hierarchy to the second level.

Yet no known polynomial time algorithm (neither classical nor quantum) for GI.

Hidden Subgroup Problem

Given: a group **G**, a finite set **S**, and a function **f** from **G** to **S** that is constant and distinct on left cosets of an unknown subgroup **H**.
Determine: Set of generators for **H**.



Hidden Subversive Problem



Solvable in constant time

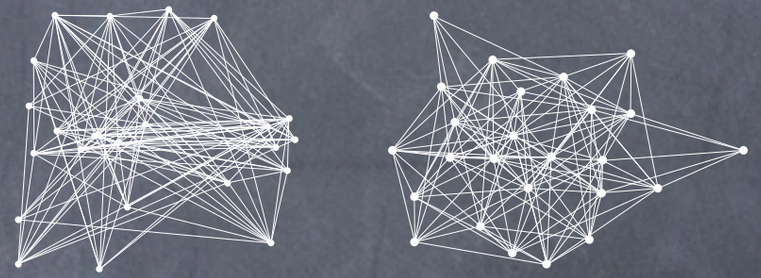
Hidden Subgroup Approach to Graph Isomorphism

Group: $G = S_{2n}$ (or $G = S_n \wr S_2$)

Function: $f(\pi) = [A_1 \oplus A_2]^\pi$

$A_i =$ adjacency matrix of graph i

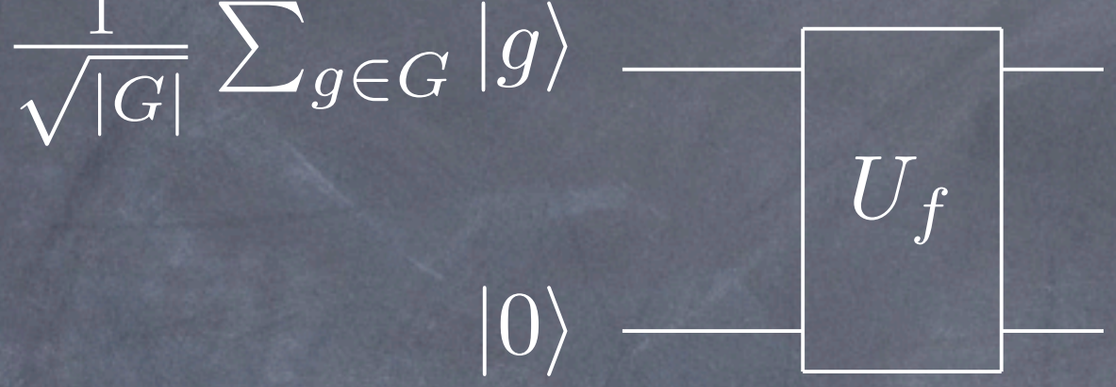
$[M^\pi]_{i,j} = [M]_{\pi(i),\pi(j)}$



$A_1 \oplus A_2$

Hidden subgroup: Graphs are isomorphic iff at least one hidden subgroup generator permutes vertices from graph 1 to graph 2

Hidden Subgroup Obstacles

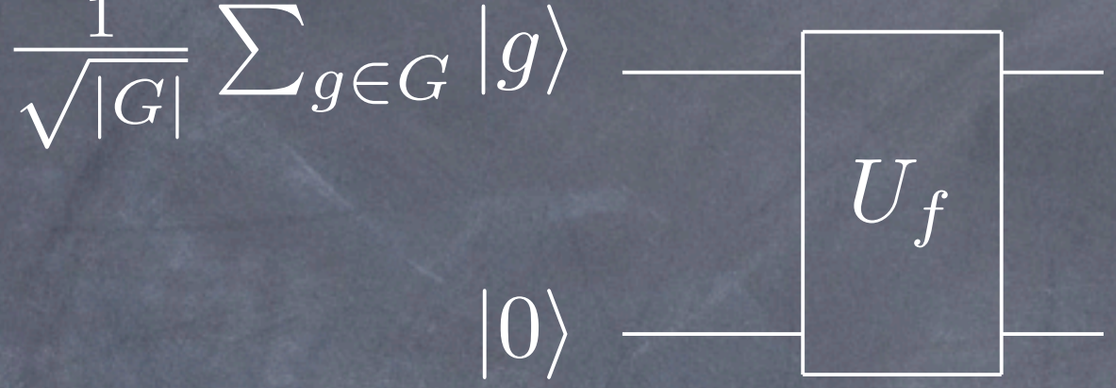


Hidden subgroup state:

$$\rho_H = \frac{|H|}{|G|}\sum_{\hat{g}\in\hat{G}}|\hat{g}H\rangle\langle\hat{g}H| \qquad |gH\rangle = \frac{1}{\sqrt{|H|}}\sum_{h\in H}|gh\rangle$$

[Hallgren, Moore, Roettler, Russell, Sen 2006]: Quantum algorithm that uses hidden subgroup states must perform measurement that are joint across O(n log n) hidden subgroup states

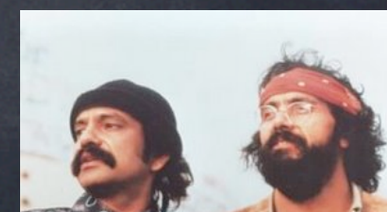
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Symmetries and GI

- Abelian HSP is good for finding hidden subgroups of large (nearly Abelian) groups (=> Shor's algorithm)
- GI is efficiently solvable when eigenvalues of the adjacency matrix are of bounded degeneracy
- Automorphism group for bounded multiplicity graphs can be trivial, polynomial size, or exponential size
- "Size" of hidden automorphism group doesn't seem to be key obstacle in efficient solution of GI

Large Object

Factoring is to Period Finding as
Graph Isomorphism is to ?

? perhaps not = HSP

Large Object



Factor

Gr

HSP

A Story - The Physicist



The eigenvalues (spectrum) of an adjacency matrix is a graph invariant.

$$\mathrm{spec}(A) = \mathrm{spec}(A^{\pi})$$

Eigenvalues are not enough to distinguish all non-isomorphic graphs (there exist non-isomorphic but iso-spectral graphs)

Terry
Rudolph

adjacency matrix => Hamiltonian of a particle

A Story - The Physicist



Terry Rudolph

One quantum particle:

$$H = A$$

A Story - The Physicist



Terry Rudolph

One quantum particle:

$$H = A$$

Two quantum particles:

$$H = A \otimes I + I \otimes A$$

A Story - The Physicist



Terry Rudolph

One quantum particle:

$$H = A$$

Two quantum particles:

$$H = A \otimes I + I \otimes A$$

Two quantum bosons:

$$H = S(A \otimes I + I \otimes A)S \quad S = \frac{1}{2}(I + SWAP)$$

A Story - The Physicist



Terry Rudolph

One quantum particle:

$$H = A$$

Two quantum particles:

$$H = A \otimes I + I \otimes A$$

Two quantum bosons:

$$H = S(A \otimes I + I \otimes A)S \quad S = \frac{1}{2}(I + SWAP)$$

Two quantum hard-core bosons:

$$H = RS(A \otimes I + I \otimes A)SR \qquad R = I - \sum_{i=1}^{n} |i,i\rangle\langle i,i|$$

A Story - The Physicist



Terry Rudolph

Two quantum hard-core bosons:

$$H = RS(A \otimes I + I \otimes A)SR$$

$$R = I - \sum_{i=1}^{n} |i,i\rangle\langle i,i|$$

[arXiv:quant-ph/0206068]

Examples of two non-isomorphic but iso-spectral graphs that when converted to above Hamiltonian have different spectra.

A Story - The Physicist



Terry Rudolph

Strongly regular graphs, however are not distinguished by two hard-core bosons spectra. Three hard-core bosons spectra did seem to work.....

See also Shiau, Joynt, Coppersmith [arXiv:quant-ph/031217]

(Note these are classical poly time algorithms motivated by quantum walks)

A Story - Some Blogger

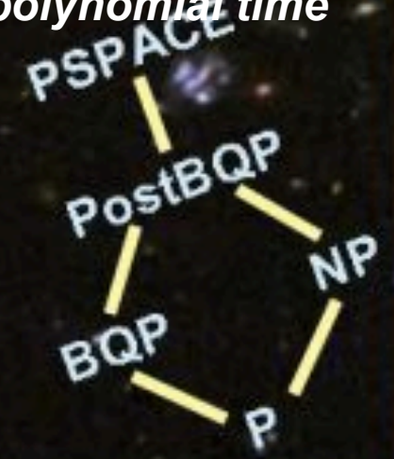
Quantum computers are not known to not solve NP-complete problems in polynomial time



Shtetl-Optimized

The Blog of Scott Aaronson

Quantum computers are not known to be able to solve NP-complete problems in polynomial time.



« Is it possible to write a competent newspaper article about math?

The ten most annoying questions in quantum computing Pipin'-hot learnin' theorems »

August 15th, 2006

A Story - The Physicist



Terry Rudolph

Tez Says:
Comment #28 August 17th, 2006 at 10:22 pm

Ok this one is annoying me:

Consider (hard core) bosons hopping around the vertices of a graph G (via a standard exchange hamiltonian). The Hamiltonian when restricted to a single particle hopping is exactly the adjacency matrix of the graph. The annoying question is this: When you have two particles hopping on either a graph G or a graph G' and the energy spectra are the same for both graphs, are the single particle spectra also the same? (i.e. are the graphs cospectral, in a graph theorists language?) Numerics and physical intuition strongly suggest so.

More difficult: Despite searching millions of likely candidate pairs of graphs, no pair of (non-isomorphic) graphs with the same spectra for 3 particles have been found. However if none such existed there would be a cubic time (classical) algorithm for graph isomorphism! I'd really like to find such a pair so I could see what is special about them.

These Hubbard model spectra are "quantum computable" graph invariants to one extent or another, so it would be nice to know something about them. What little we do know is here:

http://www.arxiv.org/abs/math.CO/0507251

A Story - The Mathematician



Greg Kuperberg

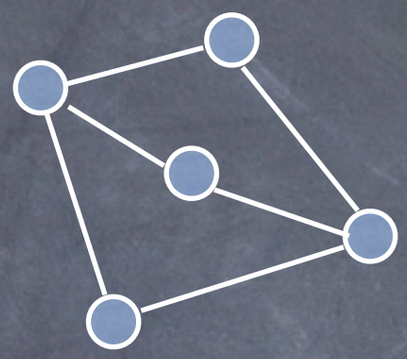
Greg Kuperberg Says:
Comment #29 August 17th, 2006 at 11:35 pm

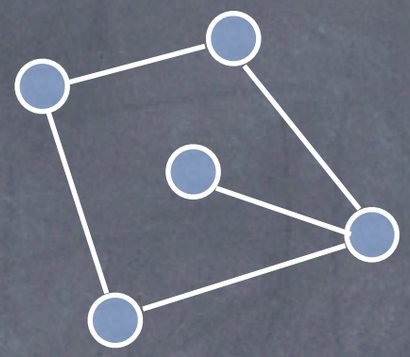
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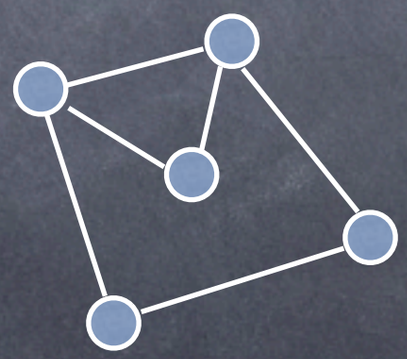
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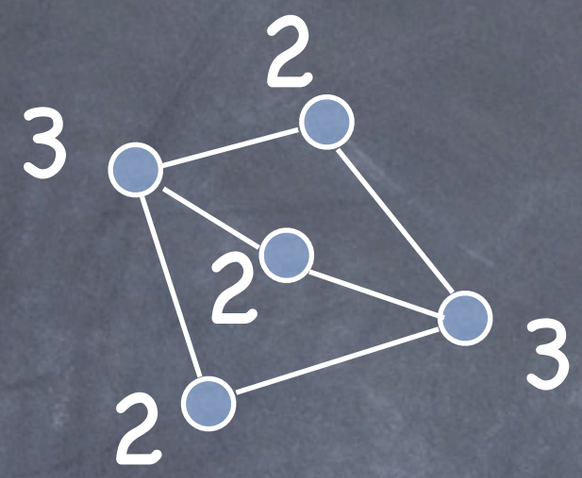
A Story - Weisfeiler-Lehman



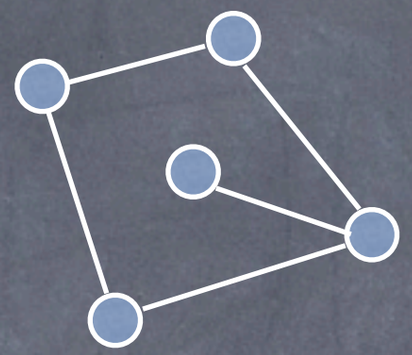


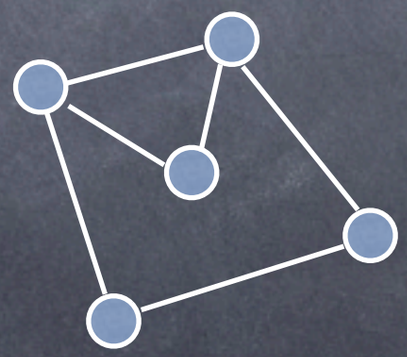


A Story - Weisfeiler-Lehman

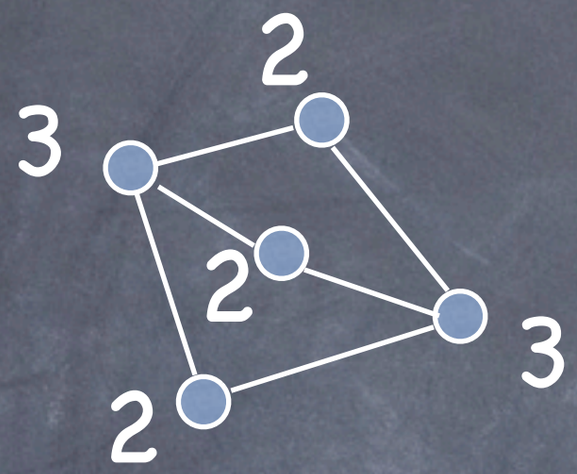


$[3^2,2^3]$

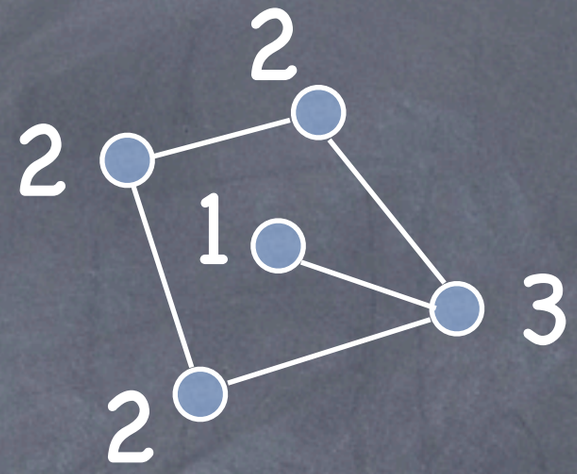




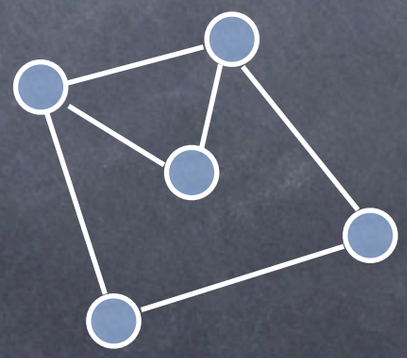
A Story - Weisfeiler-Lehman



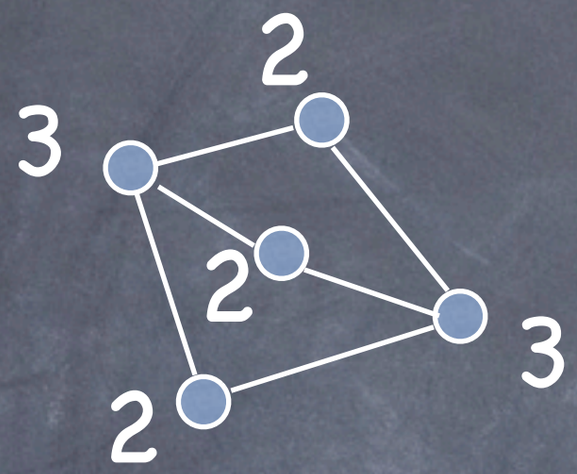
$[3^2,2^3]$



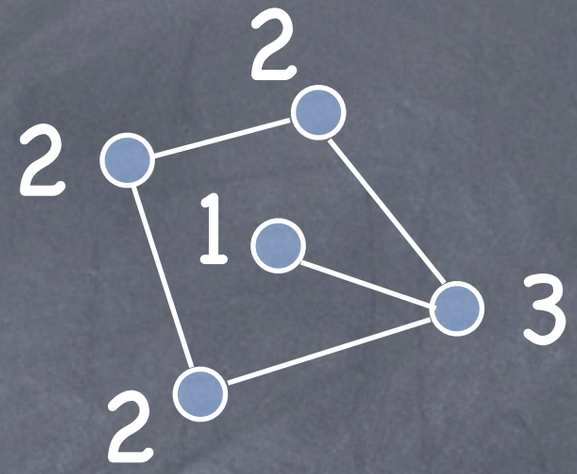
$[3,2^3,1]$



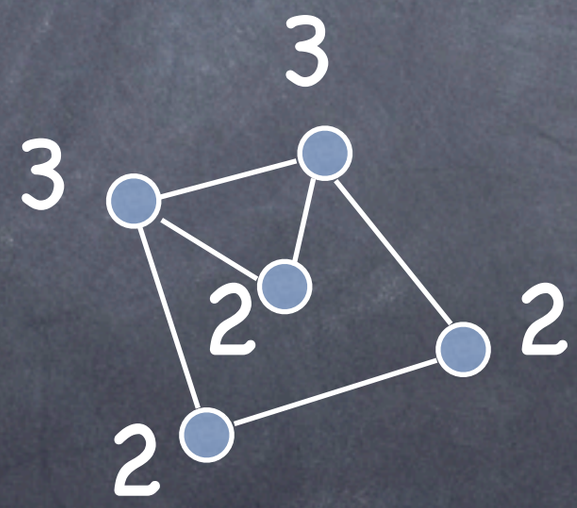
A Story - Weisfeiler-Lehman



$[3^2,2^3]$

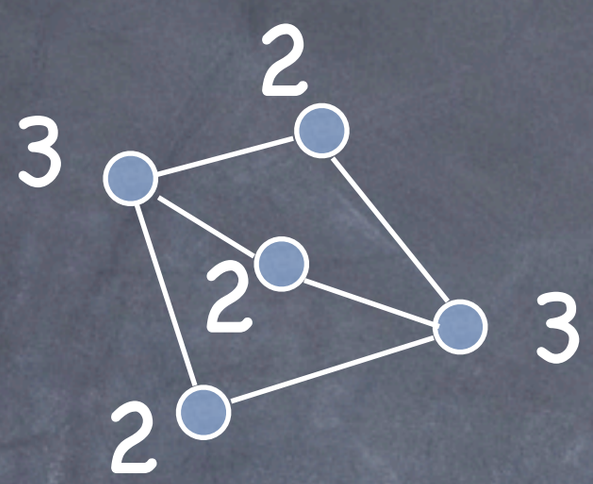


$[3,2^3,1]$

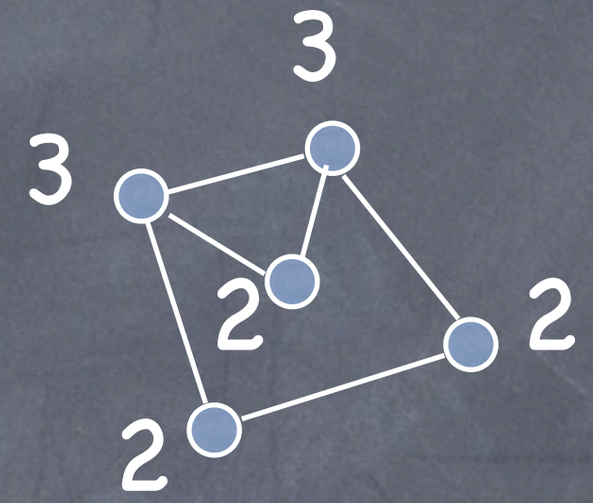


$[3^2,2^3]$

A Story - Weisfeiler-Lehman

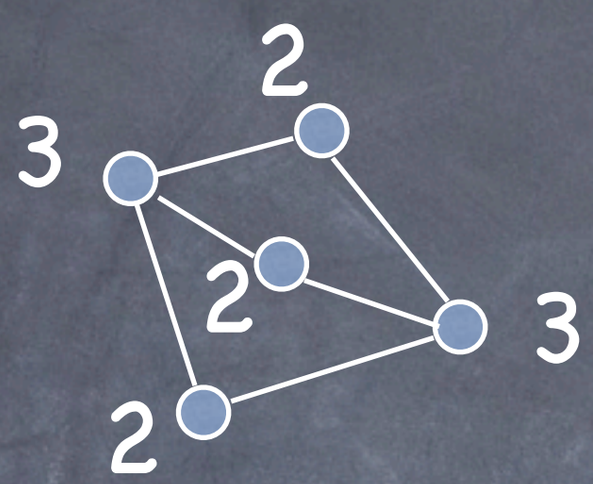


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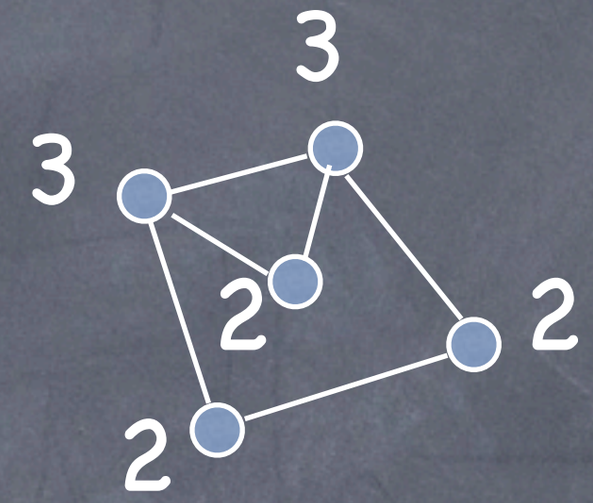


$[3^2,2^3]$

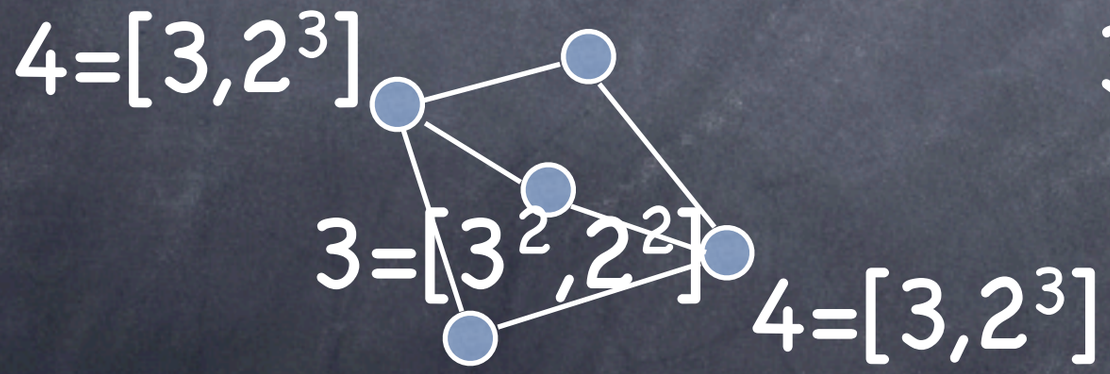
A Story - Weisfeiler-Lehman



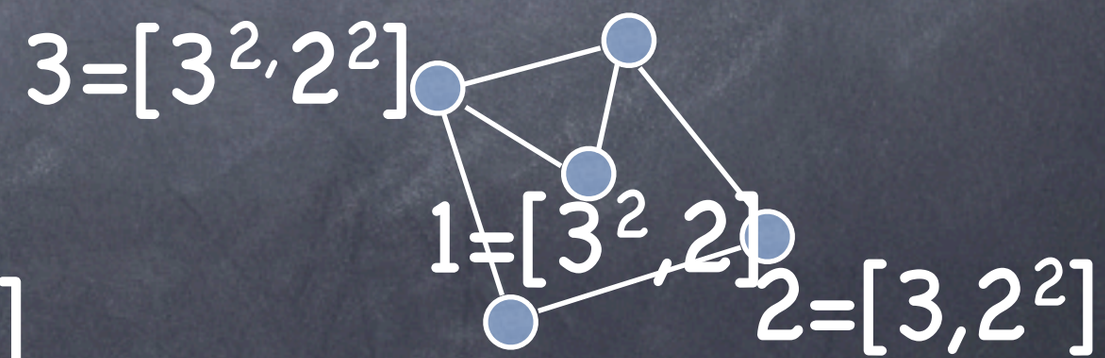
$[3^2,2^3]$



$[3^2,2^3]$



$[4^2,3^3]$



$[3^2,2^2,1]$

A Story - k-dim WL

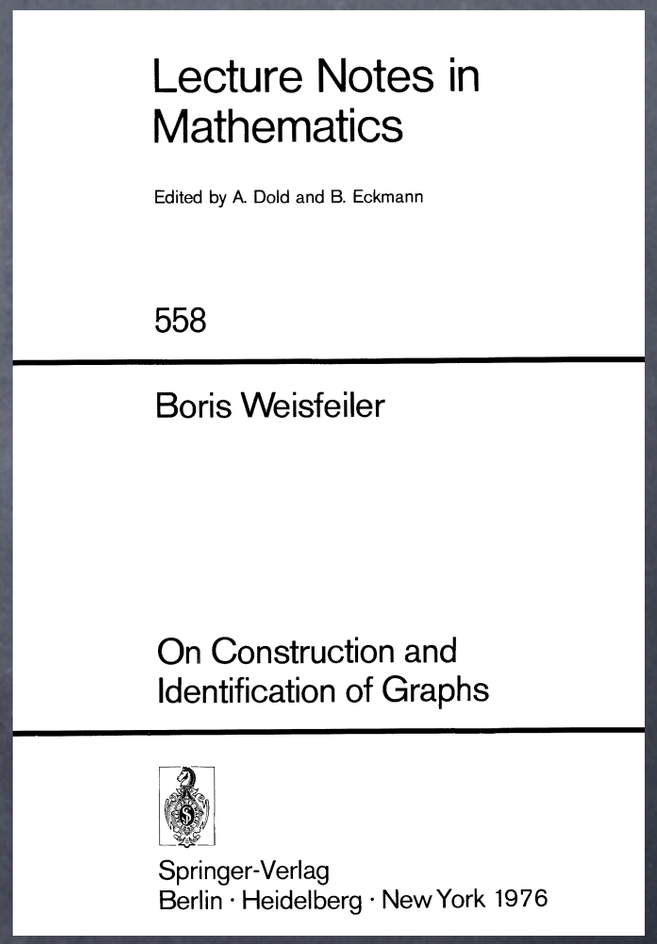


Boris Weisfeiler

Roughly corresponds to considering subsets of size k, and running a refining algorithm.

Running time n^k

Relevant mathematics: "Cellular Algebra"



Lecture Notes in Mathematics

Edited by A. Dold and B. Eckmann

558

Boris Weisfeiler

On Construction and Identification of Graphs

Springer-Verlag
Berlin · Heidelberg · New York 1976

Through 70s and 80s was not known whether always works for some fixed k (hence poly time GI algoirthm)

A Story - Cai, Furer, Immerman 1992



Jin-Yi Cai



Martin Furer



Neil Immerman

Showed explicit examples of graphs where the k-dim WL algorithm fails to distinguish between two non-isomorphic graphs for $k=\Omega(n)$

A Story - The Mathematician



Greg Kuperberg

Greg Kuperberg Says:
Comment #29 August 17th, 2006 at 11:35 pm

Tez: I don't know what you mean by the "energy spectrum" as distinct from the "single particle spectrum" of a graph.

Your question about 3 particles brings to mind an impressive result of Cai, Fürer, and Immerman (see here). It says that the graph identification strategy of coloring tuples of vertices based on their valence may require *linear*-sized tuples of vertices to get started. I am not sure, but this could imply in your case that you need a linear number of particles, not just 3. (At least, that was my intuition when I thought about these matters last year.)

A Story - The Mathematician



Greg Kuperberg

A Story - The Mathematician





Babe Ruth

arXiv.org > math > arXiv:0801.2322

Search or Article-id

Mathematics > Spectral Theory

Spectra of symmetric powers of graphs and the Weisfeiler-Lehman refinements

Alfredo Alzaga, Rodrigo Iglesias, Ricardo Pignol

(Submitted on 15 Jan 2008)

The k-th power of a n-vertex graph X is the iterated cartesian product of X with itself. The k-th symmetric power of X is the quotient graph of certain subgraph of its k-th power by the natural action of the symmetric group. It is natural to ask if the spectrum of the k-th power --or the spectrum of the k-th symmetric power-- is a complete graph invariant for small values of k, for example, for k=O(1) or k=O(log n). In this paper, we answer this question in the negative: we prove that if the well known 2k-dimensional Weisfeiler-Lehman method fails to distinguish two given graphs, then their k-th powers --and their k-th symmetric powers-- are cospectral. As it is well known, there are pairs of non-isomorphic n-vertex graphs which are not distinguished by the k-dim WL method, even for k=Omega(n). In particular, this shows that for each k, there are pairs of non-isomorphic n-vertex graphs with cospectral k-th (symmetric) powers.

k hard-core bosons don't solve graph isomorphism for same reason k-dim WL algorithm fails!

New (Quantum) Approaches to Graph Isomorphism

Cellular algebra is a subalgebra of Mat_V that is

Closed under conjugate transpose

Closed under Hadamard product

$$[A \circ B]_{i,j} = [A]_{i,j}[B]_{i,j}$$

Contains all 1s matrix J and identity matrix I

Mat_V: full matrix algebra over $\mathbb{C}$ on V

Standard Basis



Every CA has a standard basis $\mathcal{R} = \{R_i\}$

1. R_i are 0/1 matrices
2. Form a `disjoint union' of J

$$\sum_i R_i = J$$

3. $R \in \mathcal{R} \Leftrightarrow R^\dagger \in \mathcal{R}$
4. A subset of basis forms a `disjoint union' of I
5. bunch of other cool conditions

WL algorithm takes generators $\{A_1, A_2, \ldots, A_k\}$ and computes basis of generated cellular algebra.

A Tale of Two Algebras

1. Adjacency matrix of graph => cellular algebra $\langle A \rangle$

2. Permutation group $\mathcal{P}$ => centralizer algebra of $\mathcal{P}$

$$\mathcal{Z}(P) = \{A | PAP^{-1} = A, P \in \mathcal{P}\}$$

Centralizer algebra of permutation group is a cellular algebra.

Automorphism group of graph is permutation group, construct its centralizer

$$\mathcal{Z}(Aut(G)) = \{A | PAP^{-1} = A, P \in Aut(G)\}$$

Dreamland

Automorphism group of graph is permutation group, construct its centralizer (cellular algebra)

$$\mathcal{Z}(Aut(G)) = \{A | PAP^{-1} = A, P \in Aut(G)\}$$

Basis of this algebra are 1-orbits and 2-orbits of automorphism group (i.e. finding this basis would allow you to solve find orbits of vertices and hence solve GI)

DREAM: $\langle A \rangle = \mathcal{Z}(Aut(G))$ (Schurian)

Sadly: not always true

Dreamland

Automorphis utation group,
construct its ora)

$\mathcal{Z}(Aut(G$ $G)\}$

Basis of thi 2-orbits of
automorphis basis would
allow you t ces and
hence solve



DREAM: $\langle A \rangle = \mathcal{Z}(Aut(G))$ (Schurian)

Sadly: not always true

m-dim WL

Big cellular algebra:

$$\mathcal{A}^{(m)} = \langle [A]_1, [A]_2, \ldots, [A]_m, I_\Delta \rangle$$

$$[A]_i = I \otimes \cdots \otimes A \otimes \cdots \otimes I \qquad I_\Delta = \sum_{i=1}^{n} (|i\rangle\langle i|)^{\otimes m}$$

i^{th} term

basis $\mathcal{R}^{(m)} = \{R_i^{(m)}\}$

Restrict it:

$$\hat{R}^{(m)} = \{B_\Delta R_i B_\Delta^\dagger | R_i \in \mathcal{R}^{(m)}, B_\Delta R_i B_\Delta^\dagger \neq 0\}$$

$$B_\Delta = \sum_{i=1}^{n} |i\rangle\langle i|^{\otimes m}$$

cellular algebra: $\hat{\mathcal{A}}^{(m)}$

m-dim WL cont.

$$\hat{R}^{(m)} = \{B_\Delta R_i B_\Delta^\dagger | R_i \in \mathcal{R}^{(m)}, B_\Delta R_i B_\Delta^\dagger \neq 0\}$$

$$B_\Delta = \sum_{i=1}^{n} |i\rangle\langle i|^{\otimes m}$$ cellular algebra: $\hat{\mathcal{A}}^{(m)}$

$$\hat{\mathcal{A}}^{(1)} \subseteq \hat{\mathcal{A}}^{(2)} \subseteq \cdots \subseteq \hat{\mathcal{A}}^{(n)} = \mathcal{Z}(Aut(G))$$

Graph isomorphism can be solved assuming you can calculate the basis in the case of m=n.

Classical exponential complexity.
Can we get a quantum grip on this object?

A Quantum Approach

We are not afraid of objects like

$$\exp(i \sum_j c_j [A]_j + d_j I_\Delta)$$

Questions:

Hadamard product? (use max entangled states?)

Restricting to symmetric computational basis?

....

An Extreme Problem



Kids don't listen to us old dudes, work on quantum algorithms for graph isomorphism instead (statement not meant to be paradox free)

A Conspiracy Theory



MISSING IN CHILE



MATHEMATICS PROFESSOR
BORIS WEISFEILER
HAS BEEN MISSING IN
CHILE SINCE
JANUARY 4, 1985